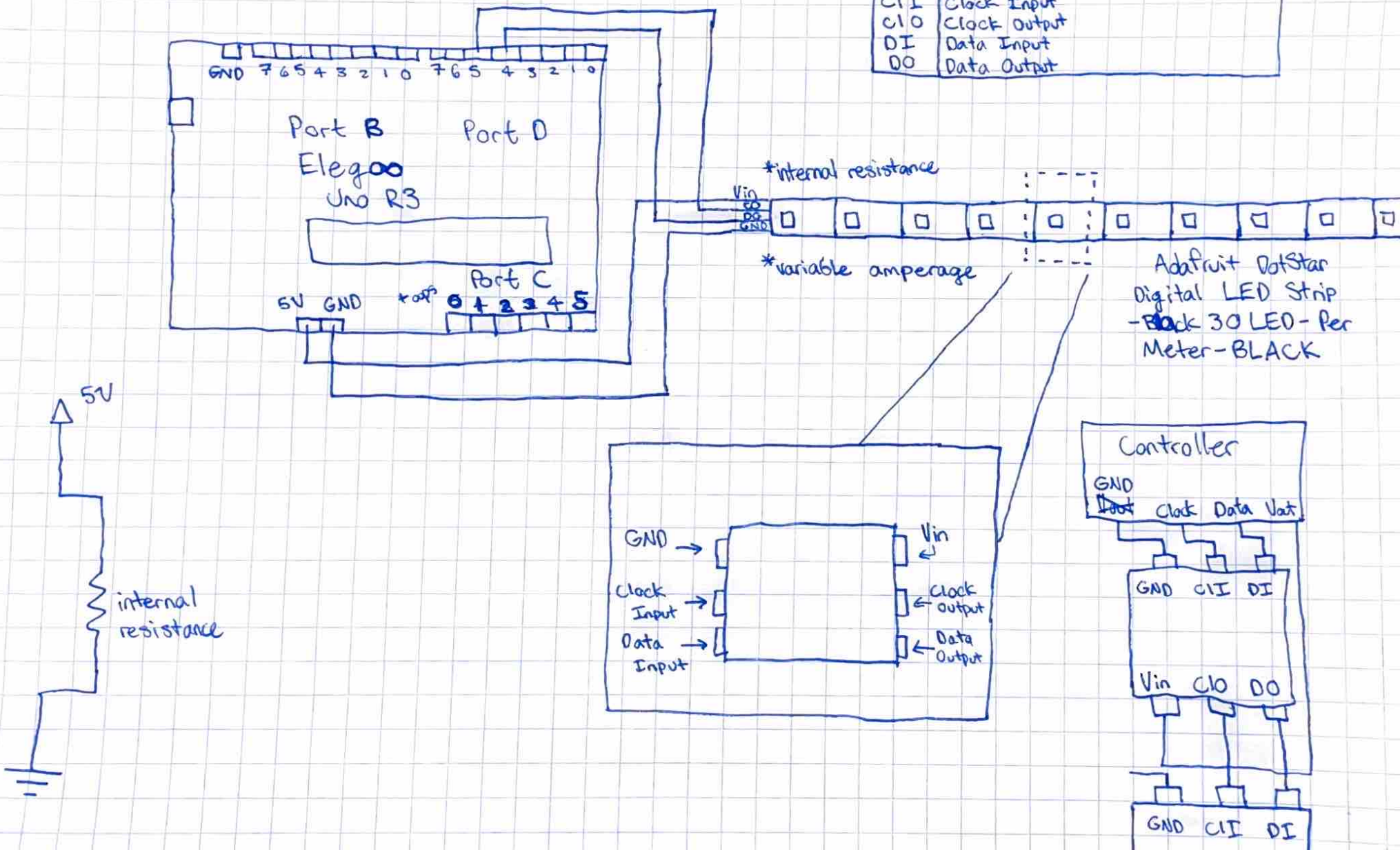
Patent Pending

Lucas Lights Datasheet

Circuit Diagrams

Key	
GND	Ground
Vin	Voltage In
Vout	Voltage Out
CI I	Clock Input
CI O	Clock Output
DI	Data Input
DO	Data Output



Memory Space Map (Initial_Animation.asm), (Lights_With_Incomplete_Input.asm)

SRAM

Red	$00 blue	$0201
	$00 green	$0202
	$FF red	$0203
	...	...
None	$00 blue	$0228
	$00 green	$0229
	$00 red	$022A

Color Input Data

- $0201 → $022A
- sets of 3 memory spaces to hold RGB data for each light color (14)
- Red → Orange → Yellow → Spring Green → Green → Turqoise → Cyan → Azure Blue → Blue → Violet → Magenta → Pink → White → None

SRAM

?	?	$0230
	...	
?	?	?

Current Sequence Data

- $0230 → ?
- Stores color data from 'Color Input Data' based on user input or current phase of animation
- L1 → L2 → ... → LN
  - for N Lights L1 → LN
- Size changes depending on # of lights

Sram

25% Brightness	$E8	$022B
	...	
(current) 25% Brightness	Starting Brightness	$022F

Brightness Data

- $022B → $022F
- Stores brightness values for LEDs
- Starting Brightness can be changed w/ USART
- 25% → 50% → 75% → 100% → Starting Brightness

Important Registers

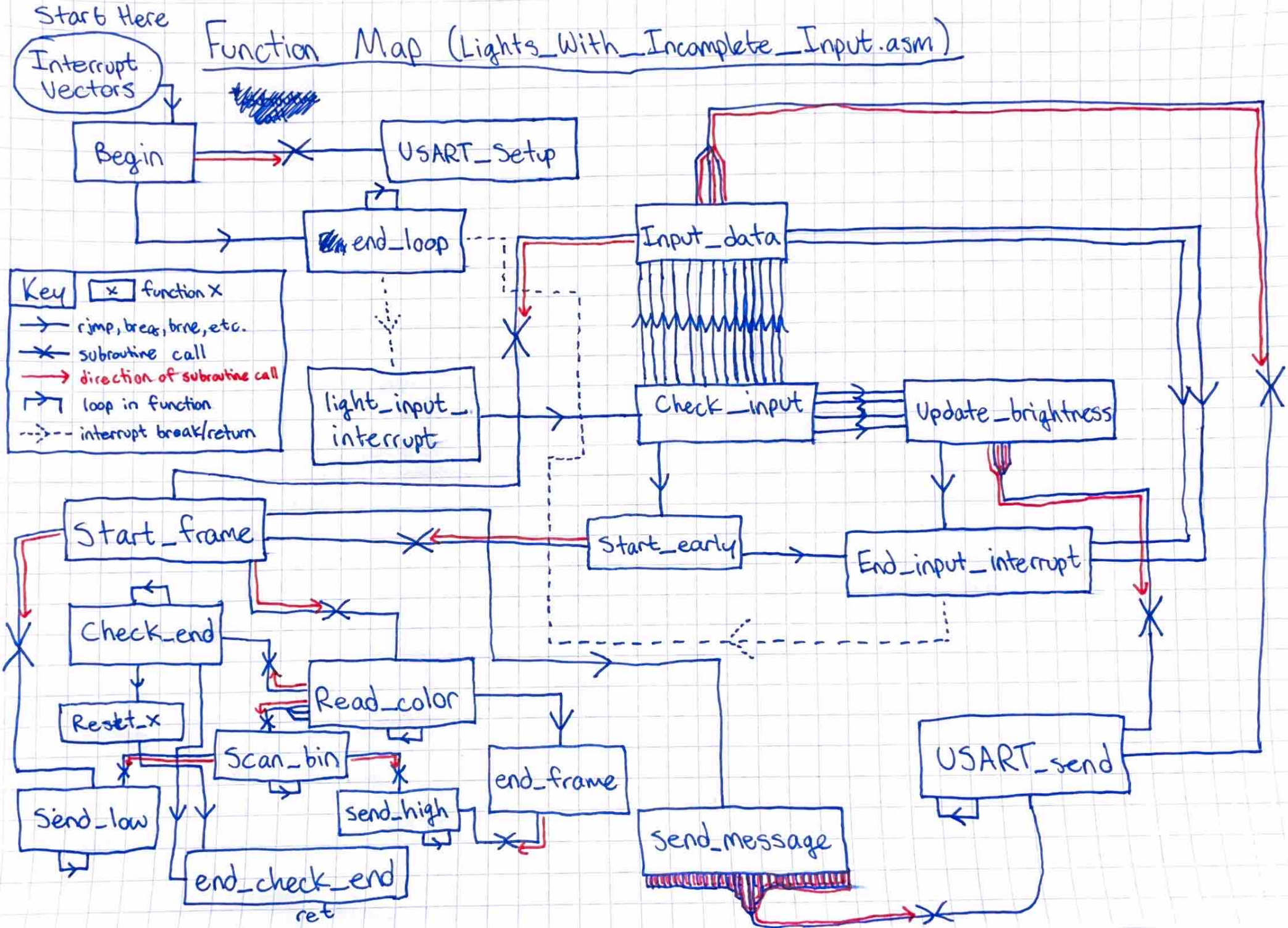
Lights w/ Incomplete Input

- r17, number of possible inputs
- r18, position offset for x pointer to next memory to add data
- r21, for loop counter
- r22, counter for # outputs

Initial Animation

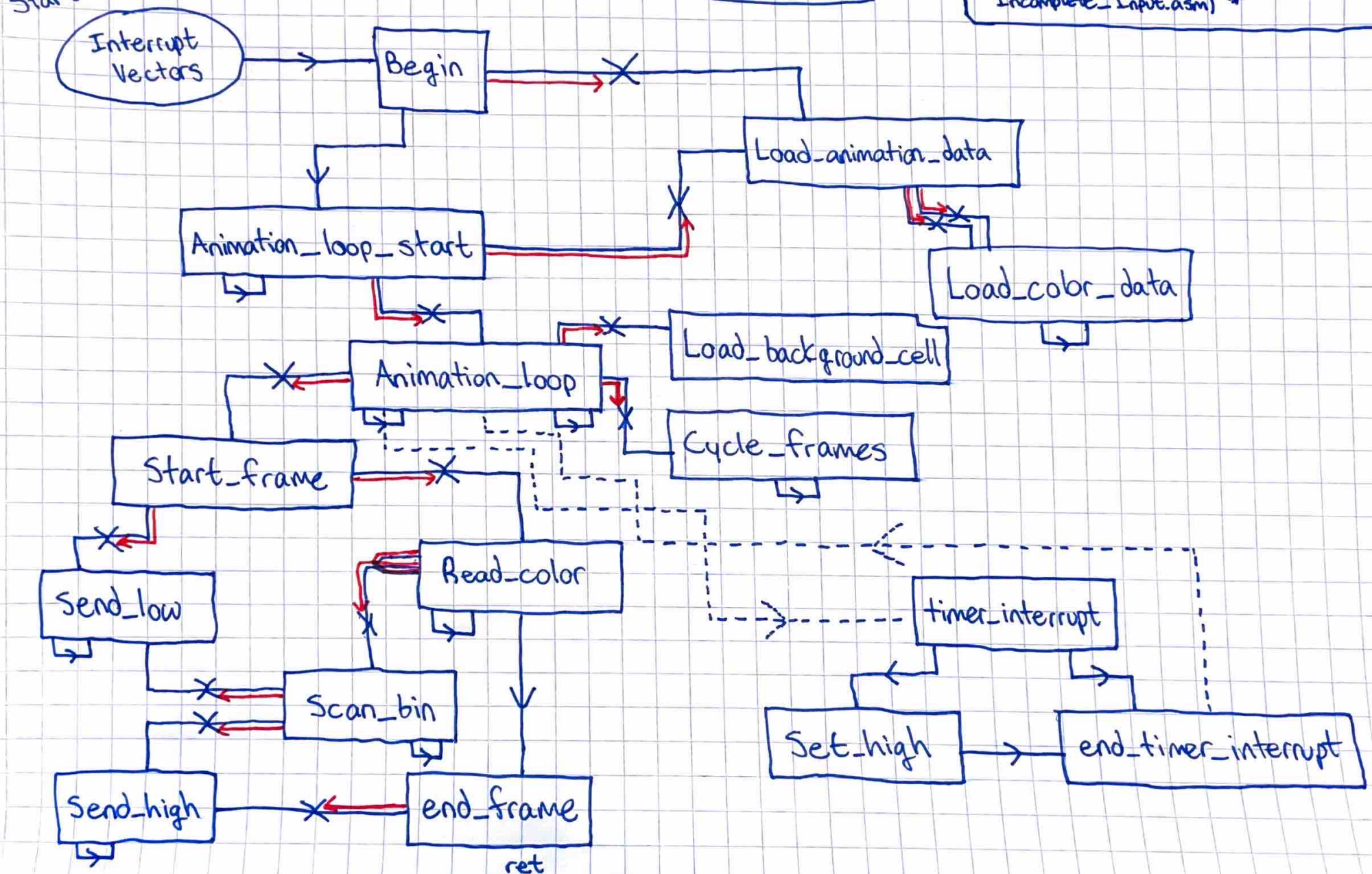
- r18, flag for timer compare
- r18, ←
- r17, number of possible inputs
- r17-19, r22-24 (in animation loop) stores color data to help shift animation

Function Map (Lights_With_Incomplete_Input.asm)



Function Map (Initial_Animation.asm)

for key see 'Function Map (Lights_With_Incomplete_Input.asm)'



Manual for Use

if Lights_With_Incomplete_Input.asm is loaded

- Program code to Elegoo Uno R3 board
- Start serial connection with putty
- Input wanted lights
  - Until desired pattern has been entered

    or

  - Until a light color has been chosen for each light in the strip
- Light Inputs ..... Desired pattern/color inputted

color	Keystroke
white	'w'
none	'n'
red	'r'
orange	'o'
yellow	'y'
spring-green	's'
green	'g'
turqoise	't'
cyan	'c'
azure	'a'
blue	'b'
violet	'v'
magenta	'm'
pink	'p'

function	keystroke
fill lights w/ pattern	'/'

- (at any point) Input Desired Brightness

Brightness Level	Keystroke
25%	'1'
50%	'2'
75%	'3'
100%	'4'

if Initial_Animation.asm is loaded

- Program code to Elegoo Uno R3 board
- Enjoy ü

Assumptions

- Kit comes pre-assembled
- Users know how to load & operate .asm files
- Users have Putty installed & know how to configure a serial connection
- Setup comes w/ 12 total lights in strip
- Setup comes w/ link to .asm, .hex, & .lst files
- Users will draw power from their computers/other USB power source